

Simultaneous Multiple Packet Capture Based on SIR Levels and Arrival Delay Offsets in CDMA Packet Networks

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Abstract—This paper proposes an analytical method to study multipacket capture effects in a code-division multiple-access (CDMA) communication system considering both signal-to-interference ratio (SIR) levels and packet arrival delay offsets. Single- and double-packet capture probabilities are derived explicitly based on CDMA cochannel interference threshold, which can be generalized to study n -packet capture effect in a CDMA packet data network. Performance of a common coded spread-spectrum (SS) packet network is taken as an example to exemplify the validity of derived analytical model for studying double-packet power-delay capture. It is concluded from the analytical results of this paper that the n -packet-capture capability can significantly enhance performance of an SS packet data network, which is usually underestimated in the literature otherwise.

Index Terms—Capture-effect, code division multiple access (CDMA), spread spectrum (SS), wireless packet network.

I. INTRODUCTION

THE PACKET capture effect in a wireless network has been a topic of interest for some time and several papers have appeared in the literature [1]–[11] to address the issue. The capture effect can be classified into two categories: power capture and delay capture. Traditionally, the study on the issue was concentrated on either the power capture (such as the papers [1]–[8], [12]) or the delay capture (such as the references [9]–[11]) separately.

The power capture occurs when received signal power of one packet overpowers the others by a certain threshold, a scenario likely exists in a packet data network due to the propagation path loss and fading. By exploiting the differences in packet power, it is possible for a receiver decode a packet successfully even if the packet is in a collision with others, or so-called “power capture” effect, which was addressed in the paper [12] by Arnbak *et al.* and the other related articles [1]–[8].

In a spread-spectrum (SS) system, when several packets encoded with the same spreading code are involved in a collision, slight difference in arrival times may give a receiver enough time to track signal correlation peaks. Once acquiring one packet, the other colliding ones encoded by either the same or different codes only contribute to multiple-access interference (MAI).

Thus a receiver is able to differentiate several colliding packets based on not only their power difference but also their distinct arrival times, as long as their arrival time offset is longer than a certain threshold. This phenomenon is called “delay capture,” which was first studied by Davis and Gronemeyer [9]. Cheun in his paper [10] derived an optimal arrival time distribution to maximize the delay capture probability. Several other papers also addressed the issue on delay capture [9], [11].

Unfortunately, the all above referred articles did not consider joint effect of power-delay capture in a code-division multiple-access (CDMA) packet network and neither they concerned about simultaneous multipacket capture (capture more than one packet) in joint power and delay capture scenario. They focused only on derivation of capture probabilities; while the analysis of a packet network with multiple packet joint power-delay capture effect was not addressed sufficiently. Thus, we are motivated in this paper to consider a slotted CDMA packet network (such as a mobile cellular system with a base station surrounded by many mobile terminals), in which a receiver can capture the first n packets from several contending ones if the spatial locations and power levels of the transmitters meet certain criteria. Considering delay and power captures jointly, we will derive a double capture probabilities (capture the first two packets in a collision). It will be shown that great benefit can be exploited from the multiple capture effect to improve the performance of an SS packet network. It should also be stressed that this paper concentrates mainly on the capture effect and its impact on the network-wise performance. We will not consider other performance-impairing factors, such as channel fading and shadowing, etc., to allow us to focus on the very fundamental issues of the capture effect. Those who are interested in the impairment to the packet capture effect caused by Rayleigh fading, shadowing and propagation path loss may refer to [16].

II. PRELIMINARIES

Before going to the details of the multiple capture study, let us first introduce some related fundamental equations.

The spatial density of packet traffic from various terminals to a certain receiver is defined as [12]

$$G_b(x) = \frac{G}{\pi} e^{-(\pi x^4/4)} \quad (0 < x < \infty) \quad (1)$$

where G is combined channel traffic and x is a random distance variable. A nonfading channel model [13] based on a flat terres-

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trial propagation model is considered in this paper, where propagation loss L varies with transmitter-to-receiver distance r as

$$L(r) = r^4 \quad (r > 0). \quad (2)$$

If all terminals transmit at a constant power P , the received power normalized by P is

$$\Gamma(r) = \frac{1}{L(r)} = r^{-4}, \quad (r > 0). \quad (3)$$

The probability distribution function (pdf) of the distance between a transmitter and a receiver is given by

$$F_R(r) = \frac{\int_0^r 2\pi x G_b(x) dx}{G} = \int_0^r 2x e^{-(\pi x^4/4)} dx. \quad (4)$$

The corresponding probability density function (pdf) is

$$f_R(r) = 2r e^{-(\pi r^4/4)} \quad (r > 0). \quad (5)$$

Let the normalized received power be $\gamma = r^{-4}$. Using variable transformation, the pdf of received power is

$$f_\Gamma(r) = \gamma^{-(3/2)} e^{-(\pi/4)\gamma} \quad (\gamma > 0). \quad (6)$$

Assume that all transmitters scatter around a receiver randomly and there are M simultaneous transmissions. The total received power is the sum of all transmitting signals power, including intended signal and interference. The pdf of the received power is calculated by M -fold convolutions of (6), or

$$f_{\Gamma_n}^{(M)}(\gamma_n) = \frac{M \gamma_n^{-(3/2)} e^{-\pi M^2/(4\gamma_n)}}{2}. \quad (7)$$

The k th user is assigned a spreading waveform $a_k(t)$ with chip duration of T_c . If $\{\alpha_j^{(k)}\}$ is the spreading sequence of elements $+1$ and -1 , $a_k(t)$ can be written as

$$a_k(t) = \sum_{j=-\infty}^{+\infty} \alpha_j^{(k)} p_{T_c}(t - jT_c) \quad (8)$$

where $p_\tau(t) = 1$ for $0 \leq t < \tau$ and $p_\tau(t) = 0$ otherwise. It is assumed that the spreading sequence, $\{\alpha_j^{(k)}\}$, has a length $N = T/T_c$ such that it covers one complete data symbol of duration T with $\alpha_0^{(k)}, \alpha_1^{(k)}, \dots, \alpha_{N-1}^{(k)}$. The data signal $b_k(t)$ can be expressed by

$$b_k(t) = \sum_{l=-\infty}^{+\infty} b_{k,l} p_T(t - lT) \quad (9)$$

where $b_{k,l} \in \{+1, -1\}$. As shown in Fig. 1, the BPSK modulated and transmitted signal for the k th user is

$$s_k(t) = \sqrt{2P_k} a_k(t) b_k(t) \cos(\omega_c t + \theta_k) \quad (10)$$

where P_k is the k th user signal power, ω_c is the carrier frequency and θ_k represents the phase of the k th carrier. The received signal $r(t)$ is given by

$$\begin{aligned} r(t) &= n(t) + \sum_{k=1}^M \lambda_k \sqrt{2P_k} a_k(t - \tau_k) b_k(t - \tau_k) \cos(\omega_c t + \varphi_k) \\ &= n(t) + \sum_{k=1}^M 2\eta_k a_k(t - \tau_k) b_k(t - \tau_k) \cos(\omega_c t + \varphi_k) \end{aligned} \quad (11)$$

where $\varphi_k = \theta_k - \omega_c \tau_k$, $n(t)$ is AWGN with the power spectral density $N_0/2$, λ_k is the path loss coefficient and $\eta_k = \lambda_k \sqrt{P_k/2}$. As relative phase shifts (modulo 2π) and relative arrival delays (modulo T) are concerned, without losing gen-

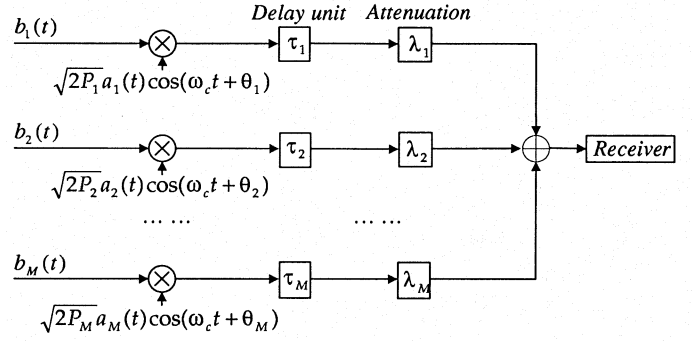


Fig. 1. The receiver model for a DS/CDMA packet data system.

erality, let us assume $\theta_i = 0$ and $\tau_i = 0$ (the i th packet is of concern) and consider only $0 \leq \tau_k < T$ and $0 \leq \theta_k < 2\pi$ for $k \neq i$.

III. CAPTURE PROBABILITIES

The packet-capture probabilities should consider both time offset (delay capture) and signal-to-interference (SIR) ratio (power capture) criteria. To capture the first packet, either of the following conditions should be met.

- The first packet has a delay offset at least T_c with respect to the second one, which is overlapped by the rest of contending packets, and the MAI is below a threshold ρ for the first packet to be captured successfully.
- The first packet has a delay offset at least T_c with respect to the second one, which has also a time offset at least T_c with respect to the rest of contending packets, and the MAI is below the threshold ρ for the first packet to be captured successfully but it exceeds the threshold for the second one to be decoded successfully (hence the second packet is swamped in the MAI, although it can be differentiated from the later ones by the time offset).

To capture the first two packets, the one condition must be met.

- The first packet has a time offset at least T_c with respect to the second one, which has also a time offset at least T_c with respect to the rest contending packets, and MAI is below SIR threshold ρ for both packets to be decoded successfully.

Similarly, the conditions for capturing more packets can also be formulated accordingly.

A. Power Capture in Terms of MAI

The SIR is defined as the power ratio of desirable signal to interference (including MAI and AWGN). For a given SIR threshold ρ , the power capture probabilities can be derived as follows. Let packet i be the intended packet. If the received signal $r(t)$ is the input to a correlating receiver i matched to $s_i(t)$, the output becomes

$$Z_i = \int_0^T r(t) a_i(t) \cos \omega_c t dt = \xi_1 + \xi_2 + \xi_3 \quad (12)$$

where the three parameters can be expressed by

$$\xi_1 = \int_0^T 2\eta_i a_i^2(t) b_i(t) \cos^2 \omega_c t dt = \eta_i b_{i,0} T, \quad (13)$$

$$\begin{aligned}\xi_2 &= \sum_{k=1, k \neq i}^M 2\eta_i \int_0^T a_k(t - \tau_k) a_i(t) b_k(t - \tau_k) \cos \omega_c t \\ &\quad \cdot \cos(\omega_c t + \varphi_k) dt \\ &= \sum_{k=1, k \neq i}^M \eta_i [b_{k,-1} R_{k,i}(\tau_k) + b_{k,0} \hat{R}_{k,i}(\tau_k)] \cos \varphi_k \quad (14)\end{aligned}$$

$$\xi_3 = \int_0^T n(t) a_i(t) \cos \omega_c t dt. \quad (15)$$

Here $R_{k,i}$ and $\hat{R}_{k,i}$ are continuous-time partial cross-correlation functions defined as (for $0 \leq \tau < T$ and $\tau = lT_c + \varepsilon$, $0 \leq \varepsilon \leq T_c$)

$$R_{k,i}(\tau) = \int_0^\tau a_k(t - \tau) a_i(t) dt \quad (16)$$

$$\hat{R}_{k,i}(\tau) = \int_\tau^T a_k(t - \tau) a_i(t) dt. \quad (17)$$

It was shown in [14], [15] that

$$\hat{R}_{k,i}(\tau) = C_{k,i}(l)T_c + [C_{k,i}(l+1) - C_{k,i}(l)](\tau - lT_c) \quad (18)$$

$$R_{k,i}(\tau) = C_{k,i}(l-N)T_c + [C_{k,i}(l+1-N) - C_{k,i}(l-N)] \cdot (\tau - lT_c) \quad (19)$$

where discrete aperiodic cross-correlation function $C_{k,i}$ for sequences $\{\alpha_j^{(k)}\}$ and $\{\alpha_j^{(i)}\}$ is defined as

$$C_{k,i}(l) = \begin{cases} \sum_{j=0}^{N-1-l} \alpha_j^{(k)} \alpha_{j+1}^{(i)} & 0 \leq l \leq N-1 \\ \sum_{j=0}^{N-1+l} \alpha_{j-1}^{(k)} \alpha_j^{(i)} & 1-N \leq l < 0 \\ 0 & |l| \geq N. \end{cases} \quad (20)$$

The SIR is calculated from $\{E[Z_i]\}^2 / \text{var}[Z_i]$. Because of symmetry, we need only consider $b_{i,0} = +1$. Conditioned on all λ_k ($k = 1, \dots, M$), the values of $\{E[Z_i]\}^2$ and $\text{var}[Z_i]$ can be calculated as

$$E[Z_i] = \eta_i T = \sqrt{P_i/2} \lambda_i T \quad (21)$$

$$\text{var}[Z_i] = \text{var}(\xi_1) + \text{var}(\xi_2) + \text{var}(\xi_3) \quad (22)$$

$$\text{var}(\xi_1) = 0 \quad (23)$$

$$\text{var}(\xi_2) = \frac{T^2}{6N^3} \sum_{k=1, k \neq i}^M \eta_k^2 [2\mu_{k,i}(0) + \mu_{k,i}(1)] \quad (24)$$

$$\text{var}(\xi_3) = \frac{N_0 T}{4} \quad (25)$$

where $\mu_{k,i}(n)$ is the cross-correlation parameter defined as

$$\mu_{k,i}(n) = \sum_{l=1-N}^{N-1} C_{k,i}(l) C_{k,i}(l+n). \quad (26)$$

Thus, if all received packets have the same power $P_k = P$, we have

$$\text{SIR} = \frac{\{E[Z_i]\}^2}{\text{var}[Z_i]}$$

$$= \frac{\frac{P}{2} \lambda_i^2 T^2}{\frac{N_0 T}{4} + \frac{T^2}{6N^3} \sum_{k=1, k \neq i}^K \frac{P}{2} \lambda_k^2 [2\mu_{k,i}(0) + \mu_{k,i}(1)]}. \quad (27)$$

We can further define

$$r_{k,i} = 2\mu_{k,i}(0) + \mu_{k,i}(1), \quad (k, i = 1, \dots, M) \quad (28)$$

$$E_b = PT \quad (29)$$

$$\Gamma_k = \lambda_k^2, \quad (k = 1, \dots, M) \quad (30)$$

where $r_{k,i}$ is the correlation parameter, E_b is the energy per data symbol and Γ_k is the normalized instantaneous power of the k th packet. Equation (27) can be rewritten into

$$\text{SIR} = \frac{\Gamma_i}{\frac{N_0}{2E_b} + \frac{1}{6N^3} \sum_{k=1, k \neq i}^M \Gamma_k r_{k,i}}. \quad (31)$$

For a set of Gold code or the least side-lobe energy/auto-optimal (LSE/AO) phased m -sequence, $r_{k,i}$ for different k has a very similar value. If let $\bar{r}$ be the average value of cross-correlation parameter in a code set, (31) can be approximated by

$$\text{SIR} = \frac{\Gamma_i}{\frac{N_0}{2E_b} + \frac{\bar{r}}{6N^3} \sum_{k=1, k \neq i}^M \Gamma_k}. \quad (32)$$

If the SIR exceeds the threshold ρ , the packet is received without errors. From (31) the probability for this event to happen is

$$\begin{aligned}\Pr\{\text{SIR} > \rho | \text{packet } i\} \\ &= \Pr\left\{\left[\Gamma_i / \left(\frac{N_0}{2E_b} + \frac{\bar{r}}{6N^3} \sum_{k=1, k \neq i}^M \Gamma_k\right)\right] > \rho\right\} \\ &= \Pr\left\{\Gamma_i > c \left(\sum_{k=1, k \neq i}^M \Gamma_k + n\right)\right\} \quad (33)\end{aligned}$$

where $c = \rho \bar{r} / (6N^3)$ is referred to as the capture ratio for a given SIR threshold and code length, and $n = (N_0/2E_b)(6N^3/\bar{r})$ is the normalized AWGN power.

Define $q_1(M)$ as the probability that SIR of the first packet exceeds the threshold ρ , $q_2(M)$ the probability that SIR of the first two packets exceeds the threshold ρ and $q_3(M)$ the probability that SIR exceeds ρ for the first packet but is lower than ρ for the second one. We obtain the following expressions:

$$q_1(M) = \Pr\left\{\Gamma_1 > c \left(\sum_{k=2}^M \Gamma_k + n\right)\right\} \quad (34)$$

$$q_2(M) = \Pr\left\{\Gamma_1 > c \left(\sum_{k=2}^M \Gamma_k + n\right), \Gamma_2 > c \left(\sum_{k=1, k \neq 2}^M \Gamma_k + n\right)\right\} \quad (35)$$

$$q_3(M) = \Pr\left\{\Gamma_1 > c \left(\sum_{k=2}^M \Gamma_k + n\right), \right.$$

$$\Gamma_2 < c \left(\sum_{k=1, k \neq 2}^M \Gamma_k + n \right) \}. \quad (36)$$

It is shown in the Appendix that

$$q_1(M) = \int_0^{1/\sqrt{cn}} e^{-(\pi x^2/4)} \operatorname{erfc} \left[\frac{(M-1)}{2} \sqrt{\frac{\pi c}{x^{-2} - cn}} \right] dx, \quad (M \geq 1) \quad (37)$$

$$\begin{aligned} q_2(M) = & \int_0^{1/\sqrt{cn/(1-c)}} e^{-(\pi x^2/4)} dx \\ & \cdot \int_x^{1/\sqrt{c(x^{-2}+n)}} e^{-(\pi y^2/4)} \\ & \cdot \operatorname{erfc} \left[\frac{(M-2)}{2} \sqrt{\frac{\pi c}{y^{-2} - c(x^{-2} + n)}} \right] dy \\ & + \int_0^{1/\sqrt{cn/(1-c)}} e^{-(\pi x^2/4)} dx \int_0^x e^{-(\pi y^2/4)} \\ & \cdot \operatorname{erfc} \left[\frac{(M-2)}{2} \sqrt{\frac{\pi c}{x^{-2} - c(y^{-2} + n)}} \right] dy, \quad (M \geq 2) \end{aligned} \quad (38)$$

$$q_3(M) = q_1(M) - q_2(M) \quad (M \geq 2). \quad (39)$$

If the values of M , c , $\text{SIR} = E_b/N_0$ and code sequence are given, (37)–(39) can be evaluated numerically.

B. Combined Power-Delay Capture

To derive joint delay-power capture probability, let us start with delay capture in a CDMA packet network.

Assume that there are $M = K + Q$ packet transmissions in CDMA channel and K of them are encoded on the same spreading code and addressed to the same receiver. The other Q packets are spread by different codes and contribute to the MAI. Let us focus on the K contending packets. When considering delay capture, we should not assume that the maximum delay is limited in only one symbol duration T , as we did in the earlier section, because of existence of different collision patterns. Let the maximum delay be T_D and the packet arrivals be a Poisson process. Hence, the arrival time of each packet is uniformly distributed in $[0, T_D]$. In general, we should consider $T_D = iT$ ($i = 1, 2, 3, \dots$). However, it can be shown that $T_D = 2T$ can be taken as a fairly good approximation. If the processing gain and the number of contending packets are large, even $T_D = T$ can be used instead.

Define $A_k(1)$ as the probability that, among K contending packets, the first one has a time offset at least T_c with respect to the second one, which is overlapped with the rest, and $A_k(2)$ as the probability that the first packet has a time offset at least T_c with respect to the second one, which keeps also a time offset at least T_c with respect to the rest. Let us consider the following two cases.

Case 1— $T_D = T$: Since the arrival times of K packets, $\omega_1, \omega_2, \dots, \omega_K$, are uniformly distributed in $[0, T]$, we have

$$A_k(1)$$

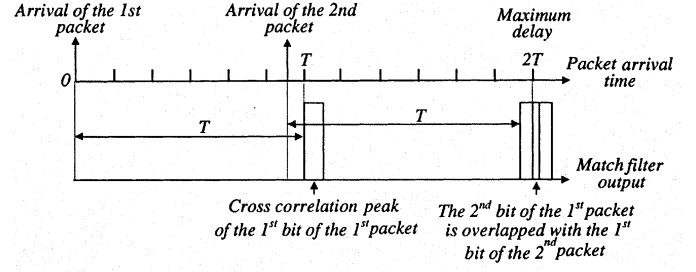


Fig. 2. A diagram to illustrate the collision scenario where the 2nd bit of the 1st packet overlaps with the 1st bit of the 2nd packet with $T_D = 2T$.

$$\begin{aligned} & = \frac{K!}{T^k} \int_0^T d\omega_1 \int_{\omega_1+T_c}^T d\omega_2 \int_{\omega_2}^T d\omega_3 \int_{\omega_3}^T d\omega_4 \cdots \int_{\omega_{k-1}}^T d\omega_k \\ & = \left(1 - \frac{1}{N}\right)^k - \left(-\frac{1}{N}\right)^k + \left(-\frac{2}{N}\right)^k - \left(1 - \frac{2}{N}\right)^k \\ & \quad + K \left(-\frac{1}{N}\right)^{k-1}, \quad (K \geq 1) \end{aligned} \quad (40)$$

where $N = T/T_c$. Similarly, we can have

$$\begin{aligned} A_k(2) & = \frac{K!}{T^k} \int_0^T d\omega_1 \int_{\omega_1+T_c}^T d\omega_2 \int_{\omega_2+T_c}^T d\omega_3 \int_{\omega_3}^T d\omega_4 \cdots \int_{\omega_{k-1}}^T d\omega_k \\ & = \left(1 - \frac{2}{N}\right)^k - \left(-\frac{2}{N}\right)^k - K \left(-\frac{1}{N}\right)^{k-1}, \quad (K \geq 1). \end{aligned} \quad (41)$$

Case 2— $T_D = iT$ ($i = 2, 3, \dots$): Now that delay may cover several symbols, collision patterns can be different from the case of $T_D = T$. A typical scenario for $T_D = 2T$ is shown in Fig. 2. The second, third packets and so on can destroy the second bit of the first packet. We will show that multiple symbol delay can be approximated by $T_D = 2T$, thus obtaining $A_k(1)$ and $A_k(2)$.

To simplify the analysis, the following mapping of the arrival times of K packets ($\omega_1, \omega_2, \dots, \omega_K$) (expressed as multiples of T) is introduced here:

$$\omega'_i = (\omega_i - \omega_1) - [\omega_i - \omega_1], \quad (i = 1, 2, \dots, K) \quad (42)$$

where $[y]$ is the largest integer that is less than y . Hence $\omega_1, \omega_2, \dots, \omega_K$ are one-to-one mapped into interval $[0, T]$. After the mapping, the first packet remains at the first position (or $\omega'_1 = 0$), while the rest are distributed randomly in the interval $[0, T]$. No matter what value T_D is, the necessary and sufficient condition for the survival of the first two packets is that they are separated from the rest by at least T_c . Hence $T_D = 2T$ is sufficient to represent any delay scenarios ($T_D = iT, i \geq 3$). To derive $A_k(1)$ and $A_k(2)$ explicitly, the interval $[0, T_D]$ is partitioned into $m = 2N$ slots, each of which has a length of T_c . Using the results given in [16, Appendix C] (for $K \geq 2$), we have

$$A_k(1) = \frac{C_1}{C} \quad (43)$$

$$A_k(2) = \frac{C_2}{C} \quad (44)$$

where

$$C_1 = \binom{m/2 + K - 3}{K - 1} - \binom{m/2 + K - 3}{K} - \binom{m + K - 5}{K} - \frac{m}{2} \binom{m/2 + K - 5}{K - 2} + \binom{m/2 + K - 5}{K} \quad (45)$$

$$C_2 = \binom{m/2 + K - 3}{K} + \binom{m + K - 5}{K} + \frac{m}{2} \binom{m/2 + K - 5}{K - 2} - \binom{m/2 + K - 5}{K} \quad (46)$$

$$C = \binom{m + K - 1}{K}. \quad (47)$$

It can be shown that, if K is large, the delay capture for $T_D = iT$ ($i \geq 2$) can be approximated by that for $T_D = T$. We use this approximation in the analysis followed. Let $Cap_{K,Q}(1)$ be the probability that receiver can capture the first packet from K contending packets if there are $M = K + Q$ packet transmissions and $Cap_{K,Q}(2)$ be the probability that the receiver can capture the first two packets. We can obtain

$$Cap_{K,Q}(1) = q_1(M)A_k(1) + q_3(M)A_k(2) \quad (48)$$

$$Cap_{K,Q}(2) = q_2(M)A_k(2). \quad (49)$$

IV. COMMON CODE NETWORK WITH DOUBLE-PACKET CAPTURE

Next, we will apply the derived capture probabilities to an SS packet network to see how much improvement in performance can be achieved.

If there are K packet transmissions in one time slot, it is just the case we discussed earlier with $M = K$ and $Q = 0$, as all packets are spread by a common code. We will show that a common code slotted SS system with double capture can significantly improve its performance. Assume that M terminals are scattered randomly around a base station with a traffic distribution given in (1) [12]. All terminals use a common code for spreading purpose. We denote single and double capture probabilities in (48) and (49) as $Cap_k(1)$ and $Cap_k(2)$ ($0 \leq K \leq M$) for $Q = 0$.

Each user can be in either “blocked” or “thinking” state. A user in the thinking state generates and transmits a new packet in a time slot with probability $P_0 = \sigma$. If the transmitted packet is not received correctly, the user enters the blocked state. A blocked user can not generate a new packet, and instead it will retransmit the old packet in the next slot with probability $P_r = p$. The probabilities σ and p are assumed to be time invariant. Let N_t (channel backlog) be the number of blocked users at time t . The one-step state transition probability for N_t

$$p_{i,j} = \Pr\{N_{t+1} = j | N_t = i\}, \quad (i, j = 0, 1, \dots, M) \quad (50)$$

takes exactly the same form as given by [16, (55) and (56)].

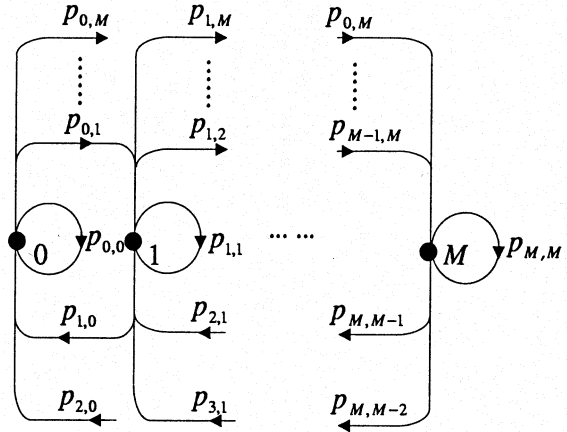


Fig. 3. The state transition diagram for a common code slotted spread spectrum packet network with double capture and $M = K, Q = 0$.

The state transition diagram is shown in Fig. 3. The steady-state occupancy probability, π_n ($n = 0, 1, \dots, M$), can be obtained from solving

$$\pi = \pi \mathbf{P} \quad (51)$$

where

$$\begin{cases} \mathbf{P} = [p_{i,j}], & (i, j = 0, 1, \dots, M) \\ \pi = (\pi_0, \pi_1, \dots, \pi_M)^T \\ \sum_{n=0}^M \pi_n = 1. \end{cases} \quad (52)$$

The conditional probability that there are exactly j ($1 \leq j \leq 2$) successful transmissions per time slot, given n blocked terminals in the system, can be expressed as

$$\begin{aligned} P(j|n) &= \Pr(j \text{ successful transmissions} | n \text{ blocked users}) \\ &= \sum_{k=0}^{M-n} \binom{M-n}{k} \sigma^k (1-\sigma)^{M-n-k} \\ &\quad \cdot \sum_{i=0}^n \binom{n}{i} p^i (1-p)^{n-i} Cap_{i+k}(j) \quad (j = 1, 2). \end{aligned} \quad (53)$$

The probability mass function of successful transmissions per time slot can be written as

$$P(i) = \sum_{n=0}^M \pi_n P(j|n) \quad (j = 1, 2). \quad (54)$$

The average number of blocked users R , the average throughput S and the normalized delay D can be defined respectively as

$$R = \sum_{n=0}^M n \pi_n \quad (55)$$

$$S = \frac{1}{2} \sum_{j=1}^2 j P(j) \quad (56)$$

$$D = \frac{R}{S} \text{ (slots)}. \quad (57)$$

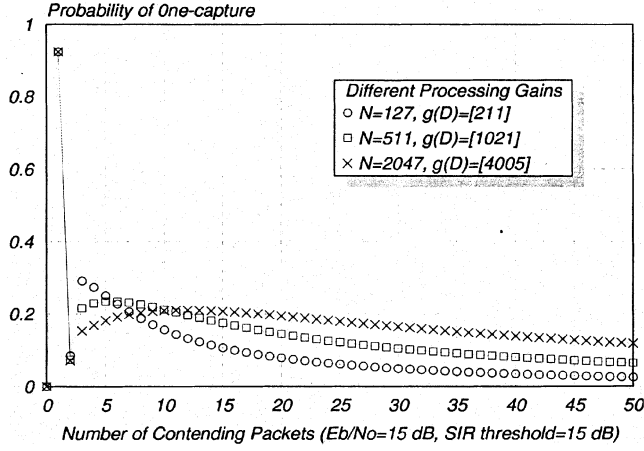


Fig. 4. $Cap_K(1)$ as a function of number of contending packets with $E_b/N_0 = 15$ dB, SIR capture threshold $\rho = 15$ dB and various code lengths N .

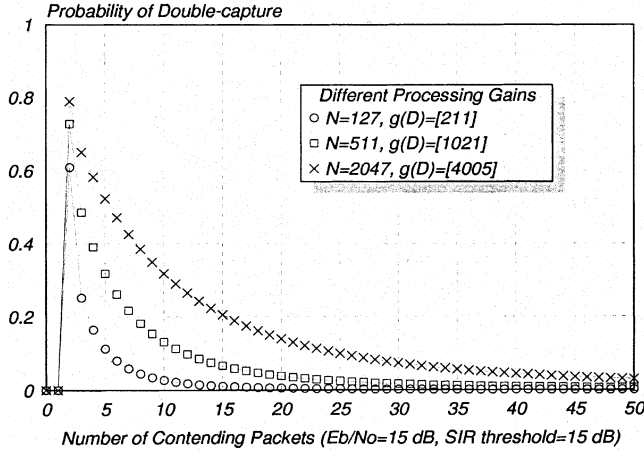


Fig. 5. $Cap_K(2)$ as a function of number of contending packets with $E_b/N_0 = 15$ dB, SIR capture threshold $\rho = 15$ dB and various code lengths N .

The channel traffic for given σ and p is defined as $G = Rp + (M - R)\sigma$.

V. NUMERICAL RESULTS AND DISCUSSIONS

The M -sequences with different lengths, $N = 127$, 511 and 2047, are taken as the spreading codes, whose generation polynomials in octal representation are [211], [1021] and [4005] respectively. Figs. 4 and 5 show single and double capture probabilities as functions of the number of contending packets, K , with different processing gains, N , where SIR threshold is chosen to be $\rho = 15$ dB. In Fig. 4, for small number of contending packets, $Cap_k(1)$ hits the lowest point when $K = 2$, because it is most likely that the double-capture can happen. As K increases, $Cap_k(1)$ begins to drop beyond certain point. It is understandable that for small K (< 2) the first packet is likely to survive even though the second one is overlapped by subsequent packets. For larger K , subsequent packets more likely overlap the first two packets and MAI can be so significant that the SIRs of the first two packets fail to exceed the required threshold. Notice in Fig. 4 that, for small K , $Cap_k(1)$ is large for small N , because, the larger N is,

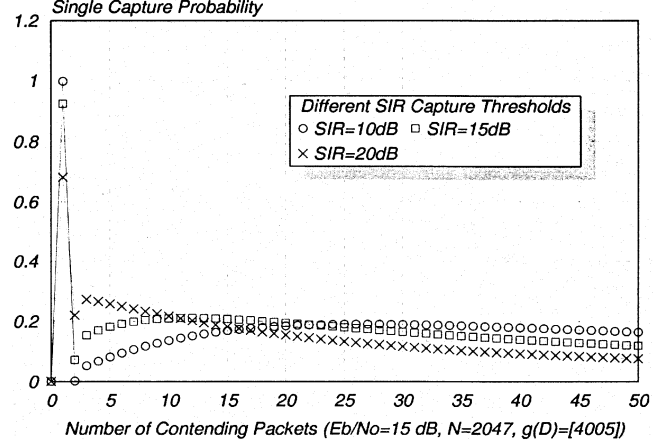


Fig. 6. $Cap_K(1)$ as a function of number of contending packets for $E_b/N_0 = 15$ dB, $N = 2047$, $g(D) = [4005]$ and different values of SIR capture threshold.

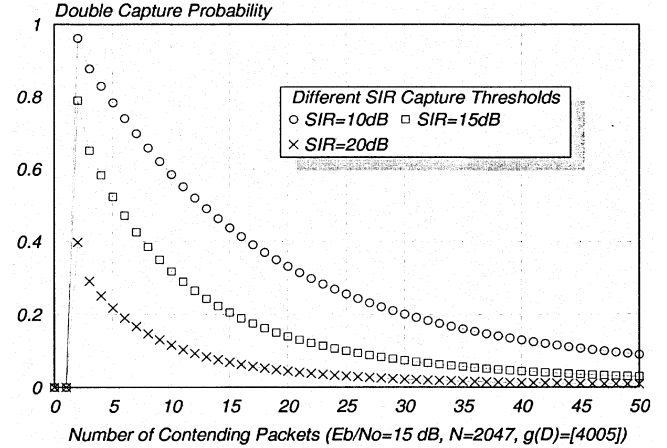


Fig. 7. $Cap_K(2)$ as a function of number of contending packets for $E_b/N_0 = 15$ dB, $N = 2047$, $g(D) = [4005]$ and different values of SIR capture threshold.

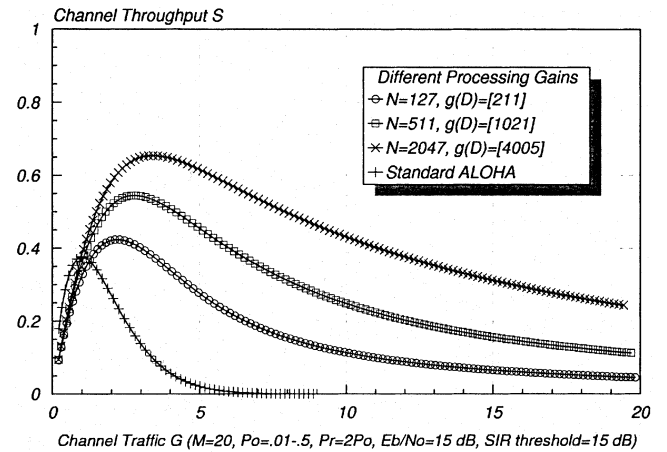


Fig. 8. Throughput as a function of channel traffic G , where $M = 20$, $P_0 = 0.01 \dots 0.5$, $P_r = 2P_0$, $E_b/N_0 = 15$ dB, SIR threshold $\rho = 15$ dB.

the more likely the both of the first two packets are captured (as shown in Fig. 5). The smaller the value of N is, the faster $Cap_k(1)$ will drop. Hence, the curve for $N = 2047$ eventually takes over the other two. Figs. 6 and 7 show the capture

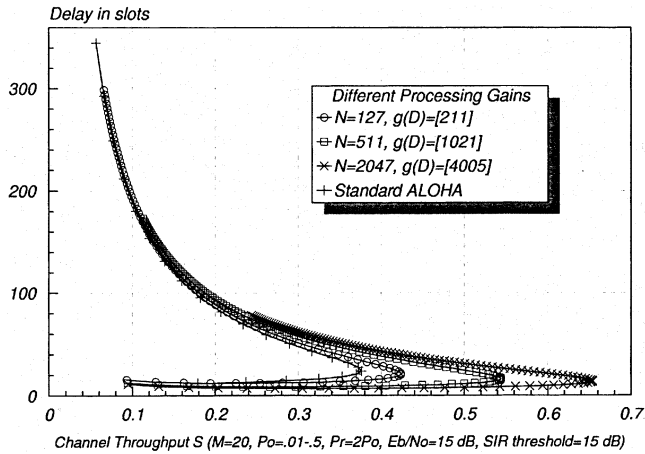


Fig. 9. Delay as a function of channel traffic G , where $M = 20$, $P_0 = 0.01 \dots 0.5$, $P_r = 2P_0$, SIR capture threshold $\rho = 15$ dB.

probability as a function of number of contending packets under different SIR threshold values, where the processing gain is $N = 2047$ with code polynomial [4005]. It is seen that a low threshold can facilitate packet-capture. The performance of a finite population (with a size of $M = 20$) SS network with double capture is shown in Figs. 8–10. Fig. 8 shows normalized throughput as a function of channel traffic. The performance for a conventional slotted ALOHA is also shown for comparison. It can be seen that the capture effect improves throughput in the network. The throughput drops more slowly as the traffic becomes higher. The larger the processing gain, the better the performance is. Fig. 9 shows the delay curves as a function of throughput, which contains a negative-slope portion, indicating the unstable region. Therefore, although the capture effect can increase the throughput, it can not eliminate collisions under heavy channel traffic. Fig. 10 shows the number of blocked users as a function of channel traffic. The steepest curve is for the standard ALOHA system, in which all users are blocked at a traffic larger than 5. On the other hand, an SS system with capture effect behaves much more nicely. For $N = 2047$, only a few terminals are blocked even at a very heavy traffic $G (> 20)$.

Fig. 11 shows the throughput as a function of traffic with different given SIR capture thresholds. Figs. 12 and 13 shows the throughput and delay as functions of SIR capture threshold, respectively. It can also be seen that, the lower the capture threshold is, the better the performance is.

It should be noted that all above performance comparisons with various spreading factors N as a parameter assume a constant data bit duration and thus the packet length. In other words, the greater the spreading factor N is used by the system, the wider the system bandwidth should be allocated. This assumption bears several advantages, one of which is that the choice of different spreading code lengths N will not affect the magnitude of offered traffic and slot size to facilitate the comparison made in this paper.

VI. CONCLUSION

This paper presents an analytical method to study the joint power and delay capture effect in a CDMA packet network. A

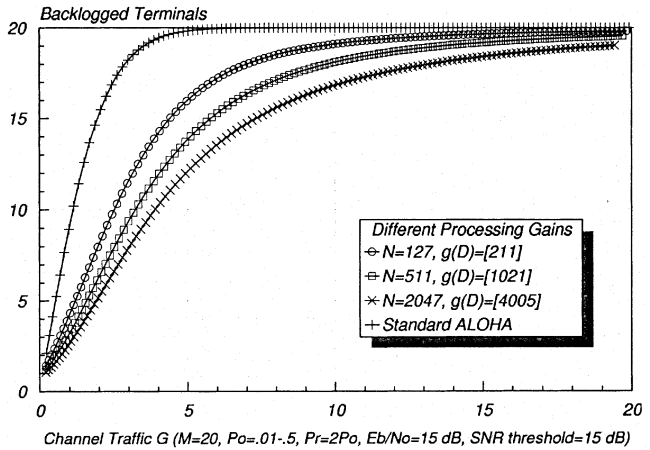


Fig. 10. Backlogged terminals as a function of channel traffic G , where $M = 20$, $P_0 = 0.01 \dots 0.5$, $P_r = 2P_0$, SIR capture threshold $\rho = 15$ dB.

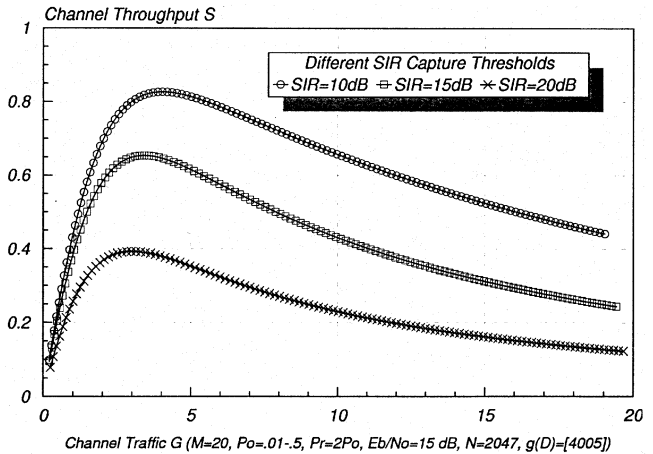


Fig. 11. Throughput as a function of channel traffic G , where $M = 20$, $P_0 = 0.01 \dots 0.5$, $P_r = 2P_0$, $N = 2047$, $g(D) = [4005]$ for different SIR capture thresholds ρ .

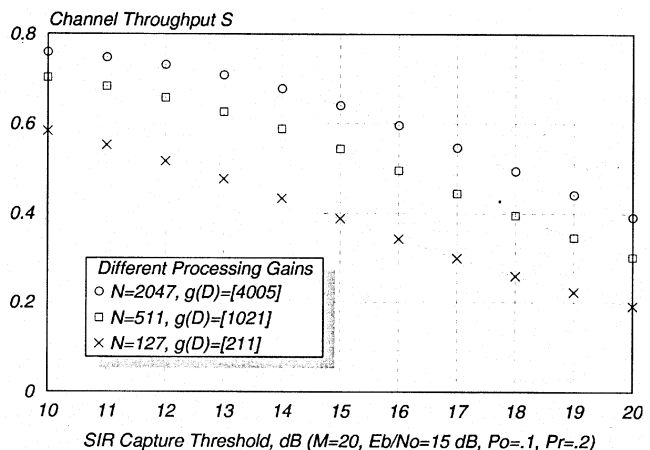


Fig. 12. Throughput as a function of SIR capture threshold ρ , where $M = 20$, $P_0 = 0.1$, $P_r = 0.2$ for different processing gains N .

closed form of double capture probabilities is derived. The analytical method can also be generalized to calculate n -packet power-delay capture ($n > 2$) probabilities. To illustrate the impact of the multiple capture effect on a packet network, a common coded SS system is explicitly studied as an example. It

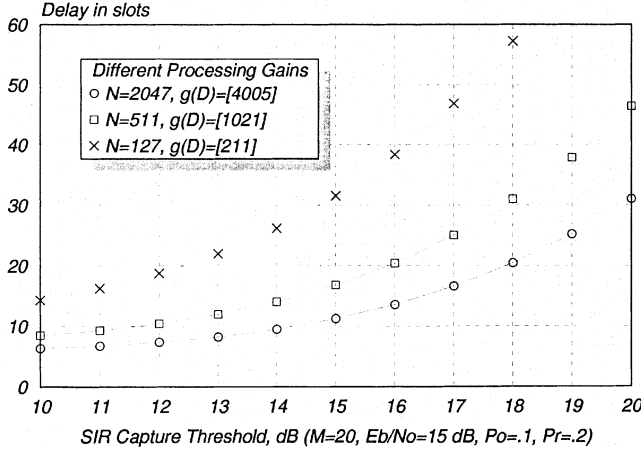


Fig. 13. Delay as a function of SIR capture threshold ρ , where $M = 20$, $P_0 = 0.1$, $P_r = 0.2$ for different processing gains N .

has been shown that the joint power-delay multiple packet capture can be preferably exploited to improve performance of a wireless packet data network.

APPENDIX DERIVATION OF POWER CAPTURE PROBABILITIES $q_1(M)$, $q_2(M)$ AND $q_3(M)$

The normalized power Γ and $\sum_i \Gamma_i$ have their distributions given in (6) and (7) respectively. Thus, we have

$$\begin{aligned}
 q_1(M) &= \Pr \left\{ \Gamma_1 > c \left(\sum_{k \neq 1}^M \Gamma_k + n \right) \right\} \\
 &= \int_0^\infty f_\Gamma(\gamma_1) d\gamma_1 \Pr \left\{ \Gamma_1 > c \left(\sum_{k \neq 1}^M \Gamma_k + n \right) \middle| \Gamma_1 = \gamma_1 \right\} \\
 &= \int_0^\infty f_\Gamma(\gamma_1) d\gamma_1 \Pr \left\{ \sum_{k \neq 1}^M \Gamma_k < \frac{\gamma_1}{c} - n \right\} \\
 &= \frac{(M-1)}{4} \int_{cn}^\infty \gamma_1^{-3/2} e^{-(\pi/4\gamma_1)} d\gamma_1 \\
 &\quad \cdot \int_0^{((\gamma_1/c)-n)} \gamma_n^{-3/2} e^{-(\pi(M-1)^2/4\gamma_n)} d\gamma_n. \quad (58)
 \end{aligned}$$

Letting $\gamma = \gamma_n^{-1/2}$, we get

$$\begin{aligned}
 q_1(M) &= \frac{(M-1)}{2} \int_{cn}^\infty \gamma_1^{-3/2} e^{-(\pi/4\gamma_1)} d\gamma_1 \\
 &\quad \cdot \int_{1/\sqrt{((\gamma_1/c)-n)}}^\infty e^{-(\pi(M-1)^2\gamma^2/4)} d\gamma \\
 &= \frac{1}{2} \int_{cn}^\infty \gamma_1^{-3/2} e^{-(\pi/4\gamma_1)} d\gamma_1 \\
 &\quad \cdot \operatorname{erfc} \left(\frac{M-1}{2} \sqrt{\frac{\pi c}{\gamma_1 - cn}} \right). \quad (59)
 \end{aligned}$$

Letting $x = \gamma_1^{-1/2}$, we have

$$q_1(M) = \int_0^{1/\sqrt{cn}} e^{-(\pi x^2/4)} \operatorname{erfc} \left(\frac{M-1}{2} \sqrt{\frac{\pi c}{x^{-2} - cn}} \right) dx. \quad (60)$$

We also can derive $q_2(M)$ as follows:

$$\begin{aligned}
 q_2(M) &= \Pr \left\{ \Gamma_1 > c \left(\sum_{k=2}^M \Gamma_k + n \right), \right. \\
 &\quad \left. \Gamma_2 > c \left(\sum_{k=1, k \neq 2}^M \Gamma_k + n \right) \right\} \\
 &= \int_0^\infty f_\Gamma(\gamma_1) d\gamma_1 \int_0^{\gamma_1} f_\Gamma(\gamma_2) d\gamma_2 \\
 &\quad \cdot \Pr \left\{ \sum_{k=3}^M \Gamma_k < \frac{\gamma_2}{c} - (\gamma_1 + n) \right\} \\
 &\quad + \int_0^\infty f_\Gamma(\gamma_1) d\gamma_1 \int_{\gamma_1}^\infty f_\Gamma(\gamma_2) d\gamma_2 \\
 &\quad \cdot \Pr \left\{ \sum_{k=3}^M \Gamma_k < \frac{\gamma_2}{c} - (\gamma_1 + n) \right\}. \quad (61)
 \end{aligned}$$

Let $x = \gamma_1^{-1/2}$, $y = \gamma_2^{-1/2}$, $\gamma = \gamma_n^{-1/2}$. We can simplify (31) into:

$$\begin{aligned}
 q_2(M) &= \int_0^{1/\sqrt{cn/(1-c)}} e^{-(\pi x^2/4)} dx \\
 &\quad \cdot \int_x^{1/\sqrt{c(x^{-2}+n)}} e^{-(\pi y^2/4)} dy \\
 &\quad \cdot \operatorname{erfc} \left[\frac{(M-2)}{2} \sqrt{\frac{\pi c}{y^{-2} - c(x^{-2} + n)}} \right] dy \\
 &\quad + \int_0^{1/\sqrt{cn/(1-c)}} e^{-(\pi x^2/4)} dx \int_0^x e^{-(\pi y^2/4)} dy \\
 &\quad \cdot \operatorname{erfc} \left[\frac{(M-2)}{2} \sqrt{\frac{\pi c}{x^{-2} - c(y^{-2} + n)}} \right] dy. \quad (62)
 \end{aligned}$$

Thus, based on $q_1(M)$ and $q_2(M)$, $q_3(M)$ ($M \geq 2$) can be evaluated using (39).

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